elenchus.ai -- 1-Page Solution Pitch

Live Application: https://elenchus-ai.indiasouthcentral.cloudapp.azure.com
GitHub Repository: https://github.com/akumar015/elenchus.ai
Competition Track: Gnani AI Voice Engineering Challenge -- Indic Dialectic Voice Agent

1. THE ONE-LINER

> elenchus.ai is a real-time, voice-first Socratic companion that cures passive information consumption by transforming philosophy, essays, and daily news into active verbal sparring sessions--forcing you to defend your logic out loud while auto-capturing your insights into permanent, structured Atomic Thought Cards.

2. THE PROBLEM: THE 'ILLUSION OF COMPETENCE'

Millions of people consume hours of podcasts, audiobooks, and AI audio overviews (like NotebookLM) every week, yet retain almost none of it.

- **[*] Passive listening is intellectual junk food:** Listening to someone explain game theory or cognitive bias feels productive, b learning. Without active recall or friction, the illusion of mastery evaporates the moment you try to explain it.
- **[*] The 'NPR Spectator Trap':** Existing AI podcast generators treat the user as a passive bystander listening to synthetic hosts banter. There is zero pushback, zero personal accountability, and zero active engagement.
- **[*] Lost Epiphanies:** Genuine realizations during a commute or walk vanish because hands are busy and note-taking has high friction.

3. THE SOLUTION: CONVERSATIONAL DIALECTIC UNDER COGNITIVE FRICTION

elenchus.ai flips the paradigm from passive broadcast to active Socratic cross-examination across four pillars:

1. **Dual-Modality Conversational Architecture:**
   - **Interactive Podcast Studio (3-Way):** Two AI co-hosts (Kalpana/Shivani Anchor + Avinash Provocateur) clash in real-time, unpack case studies from in-context markdown dossiers, and pull you into the discussion as an equal peer.
   - **Socratic Duel (1-on-1 Arena):** Adversarial cross-examination targeting fallacies across 4 rigor stages (Mentor, Balanced, Expert, Hardc Zero tolerance for unbacked premises; demands precise operational definitions.
2. **Native Indic Multilingual Code-Switching:**
   Powered by Gnani Timbre v3.0 (24kHz TTS) & Prisma STT. Flawlessly code-switches across English (en-IN), Hinglish (hi-IN), Punjabi (pa-IN), and Telugu (te-IN) with natural conversational cadence and authentic Indic phonology.
3. **Decoupled Low-Latency Voice Pipeline:**
   Overcomes speech synthesis latency by synchronously synthesizing the opening anchor beat while background-scheduling follower host segments, achieving sub-1.5s time-to-first-audio without dead-air.
4. **The Ambient Scribe & Atomic Thought Cards:**
   An ambient scribe analyzes your arguments, producing an Atomic Thought Card: isolating the objective core thesis, your articulated stance, exposed blind spots, and key insights exportable directly to Notion/Obsidian.

4. TECHNICAL STACK & PRODUCTION DEPLOYMENT

- **[*] AI Cognitive Engine:** Google Gemini 3.5 Flash Lite (Primary) with resilient fallback to Gemini 2.5 Flash & 2.0 Flash (structured JSON sc
- **[*] Voice Intelligence:** Gnani Timbre v3.0 (24kHz neural synthesis) & Gnani Prisma STT (16kHz chunked transcription).
- **[*] Backend Architecture:** FastAPI (Python 3.11), SSE streaming, asynchronous concurrency management, in-context topic dossiers.
- **[*] Frontend Application:** Next.js 14 (React, Tailwind CSS, TypeScript), Swiss archival aesthetic, full-duplex audio recorder.
- **[*] Cloud Infrastructure:** Azure Linux VM (Ubuntu 22.04), Caddy automated Let's Encrypt SSL reverse proxy, Docker Compose.

5. WHY IT WINS

Unlike passive audio summaries that tell you what to think, elenchus.ai teaches you how to think. By marrying cutting-edge Indic voice synthesis with Socratic friction, it turns idle listening into permanent cognitive fitness.